

Planning & Reinforcement Learning

Artificial Intelligence

Jacopo Parretti

I Semester 2025-2026

Indice

I	Classical Planning	4
1	Introduction to Planning	4
1.1	Historical Context and Motivation	4
1.2	The Planning Cycle	4
1.2.1	Challenges in Real-World Planning	5
2	Formalization of the Planning Problem	6
2.1	State Transition Systems	6
2.1.1	Properties of State Transition Systems	6
2.2	Classical Planning: Fundamental Assumptions	6
2.2.1	Relaxing Classical Assumptions	7
2.3	The Planning Problem	8
2.3.1	Goal Specification	8
2.3.2	Plan Quality	8
3	Representation of Planning Problems	9
3.1	The Challenge of Explicit Representation	9
3.2	Action Schemas: Structured Representation	9
3.2.1	Advantages of Action Schemas	9
3.3	State Representation: Propositional Logic	9
3.3.1	State Update Semantics	10
3.4	Example: Blocks World	11
3.4.1	Domain Description	11
3.4.2	Action Schemas	11
3.4.3	Example Problem Instance	12
II	Advanced Planning Representations	13
4	Reduction to Propositional Satisfiability	13
4.1	Motivation	13
4.2	Example Domain: Dock Worker Robots (DWR)	13
4.2.1	Domain Description	13
4.2.2	Initial State Configuration	13
4.3	Formal Representation	14
4.3.1	Objects and Types	14
4.3.2	Static Relations	14
4.3.3	Rigid Relations	15
4.4	State Representation in DWR	15
4.4.1	What is a State?	15
4.4.2	State as Truth Assignment	15
4.5	Two Representation Formalisms	16
4.5.1	Classical (Propositional) Representation	16
4.5.2	State Variable Representation	16
4.5.3	Example: Complete State Representation	17
4.6	Key Insight: States as Boolean Assignments	17
5	Action Schemas in DWR	17
5.1	Structure of Action Schemas	17

5.2	Example: Take/Load Action	18
5.2.1	Correcting Static Relations	18
5.2.2	Preconditions	18
5.2.3	Effects	19
5.3	Action Applicability	19
5.3.1	Examples	19
6	State Transition Function	20
6.1	Formal Definition	20
6.2	Computing the Successor State	20
6.3	Goal States	20
6.4	Comparing Representations: Simplified DWR Example	20
6.4.1	Classical Representation	21
6.4.2	State Variable Representation	21
6.4.3	Key Differences	21
7	Planning as Satisfiability (SAT)	21
7.1	Recap: Core Concepts	21
7.2	Temporal Instantiation of Variables	22
7.3	The Bounded Planning Problem	22
7.4	SAT Encoding	22
7.5	Solving Planning via Iterative Deepening	22
III	SAT Solvers and Planning	24
8	The Boolean Satisfiability Problem	24
8.1	Definition and Scope	24
8.2	Propositional Logic Basics	24
8.3	Interpretations and Truth Assignments	24
8.3.1	Satisfiability of Clauses	24
8.4	SAT Decision Procedure	25
8.5	The Empty Clause	25
9	Planning Problem Formalization	25
9.1	States as Interpretations	25
10	Bounded Planning Problem	26
10.1	Iterative Deepening on Plan Length	26
10.2	Temporal Encoding of Literals	26
10.2.1	Example: Temporal Instantiation	26
10.3	Encoding the Initial State	26
10.3.1	From Literals to Propositional Variables	27
10.3.2	Example: Complete Initial State Encoding	27
10.4	Encoding the Goal States	27
10.5	Encoding Actions	27
10.5.1	Action Preconditions	27
10.5.2	Mutual Exclusion of Actions	28
10.6	Complete SAT Encoding	28
10.7	Plan Extraction from SAT Model	28

Parte I

Classical Planning

1 Introduction to Planning

1.1 Historical Context and Motivation

Planning has been a central topic in artificial intelligence since the inception of the field at the **Dartmouth Conference in 1956**, where the founding fathers of AI first gathered to define the scope and ambitions of this new discipline. Planning represents one of the fundamental capabilities required for intelligent behavior: the ability to reason about sequences of actions that achieve desired goals.

The motivations for studying automated planning are manifold:

- **Autonomous agents:** Enabling robots, software agents, and autonomous systems to make decisions and act independently in complex environments
- **Resource optimization:** Finding optimal or near-optimal strategies for resource allocation, scheduling, and logistics
- **Scientific applications:** Modeling and solving problems in domains such as space exploration, manufacturing, and healthcare
- **Theoretical foundations:** Understanding the computational complexity and formal properties of reasoning about action and change

1.2 The Planning Cycle

The complete planning process involves several interconnected phases that form a continuous cycle:

1. **Plan Generation:** Given a description of the current state, available actions, and desired goals, synthesize a sequence of actions (a plan) that transforms the initial state into a goal state. This phase involves:
 - Analyzing the initial state to understand what is currently true
 - Identifying the gap between the current state and the goal state
 - Searching through the space of possible action sequences
 - Evaluating candidate plans for correctness and optimality
 - Selecting the best plan according to specified criteria (e.g., shortest length, minimum cost, fastest execution)
2. **Plan Deployment:** Execute the generated plan in the actual environment, monitoring the execution to detect deviations from expected behavior. This phase includes:
 - Translating abstract plan actions into concrete executable commands
 - Monitoring sensors and feedback to verify that actions have the expected effects
 - Detecting discrepancies between predicted and actual states
 - Maintaining a record of executed actions for debugging and learning

3. **Replanning:** When execution monitoring detects that the current plan is no longer valid (due to unexpected events, action failures, or environmental changes), generate a new plan or repair the existing one. Replanning strategies include:

- **Plan repair:** Modify the existing plan minimally to accommodate the new situation
- **Replan from scratch:** Generate an entirely new plan from the current state
- **Contingency planning:** Use pre-computed alternative plans for anticipated failures

This cycle reflects the reality that planning systems must operate in dynamic, partially unpredictable environments where initial plans may need adaptation. The cycle continues iteratively until the goal is achieved or deemed unreachable.

1.2.1 Challenges in Real-World Planning

Real-world deployment of planning systems faces several challenges:

- **Execution uncertainty:** Actions may fail or have unexpected outcomes
- **Incomplete information:** The planner may not have complete knowledge of the environment
- **Dynamic environments:** The world may change while the plan is being executed
- **Computational constraints:** Planning must often occur in real-time with limited computational resources
- **Plan quality vs. planning time:** Trade-off between finding optimal plans and responding quickly

These challenges motivate the development of robust planning algorithms that can handle uncertainty, adapt to changes, and operate efficiently under resource constraints.

2 Formalization of the Planning Problem

2.1 State Transition Systems

The mathematical foundation of planning rests on the concept of a **state transition system**, which provides a formal model of how actions transform states.

Definizione 2.1 (State Transition System). A state transition system is a tuple $\Sigma = \langle S, A, \gamma \rangle$ where:

- S is a finite or countably infinite set of **states**
- A is a finite or countably infinite set of **actions**
- $\gamma : S \times A \rightarrow S$ is a **state transition function** that maps a state and an action to a resulting state

The transition function $\gamma(s, a) = s'$ specifies that executing action a in state s results in state s' . This function encodes the **dynamics** of the domain—how the world changes in response to actions.

2.1.1 Properties of State Transition Systems

State transition systems can be characterized by several important properties:

- **Determinism:** In a deterministic system, γ is a function—each state-action pair leads to exactly one successor state. In non-deterministic systems, multiple outcomes are possible.
- **Reachability:** A state s' is reachable from state s if there exists a sequence of actions that transforms s into s' . The reachable state space from an initial state is often much smaller than the total state space.
- **Reversibility:** An action is reversible if there exists another action that undoes its effects. Many real-world actions are irreversible (e.g., breaking an object).
- **State space structure:** The connectivity and topology of the state space significantly affect planning complexity. Highly connected spaces may have many solution paths, while sparse spaces may have few or no solutions.

2.2 Classical Planning: Fundamental Assumptions

Classical planning makes several simplifying assumptions that restrict the class of problems considered but enable efficient algorithmic solutions. These assumptions define the **classical planning framework**:

1. **Finitely many states:** The state space S is finite, allowing exhaustive search techniques. This assumption ensures that the planning problem is decidable and that search algorithms will terminate.

Justification: While real-world domains may have infinite state spaces (e.g., continuous variables), finite approximations are often sufficient for practical purposes.

2. **Finitely many actions:** The action space A is finite, ensuring decidability. Each state has a finite branching factor in the state space graph.

Justification: Even in complex domains, the number of distinct action types is typically manageable, though the number of ground actions (instantiated with specific objects) may be large.

3. **Deterministic transition function:** For every state s and action a , there is exactly one resulting state $\gamma(s, a)$. Actions have predictable, certain outcomes.

Justification: Determinism simplifies reasoning about action effects. Non-deterministic planning requires more complex formalisms (e.g., Markov Decision Processes).

4. **Full observability:** The planner has complete knowledge of the current state at all times. There is no uncertainty about which state the system occupies.

Justification: Full observability eliminates the need for belief state reasoning and sensing actions. Partial observability requires more sophisticated planning approaches.

5. **Instantaneous actions:** Actions have no duration; they occur instantaneously. Temporal reasoning is not required.

Justification: Ignoring action durations simplifies the planning model. Temporal planning extends classical planning to handle durations and concurrent actions.

6. **No exogenous events:** Only the planning agent can change the world through deliberate actions. The environment remains static unless acted upon.

Justification: Exogenous events (e.g., other agents, natural processes) introduce additional complexity. Classical planning assumes a static world between actions.

These assumptions, while restrictive, capture a significant and important class of planning problems and provide a foundation for understanding more complex, realistic planning scenarios.

2.2.1 Relaxing Classical Assumptions

Modern planning research has developed extensions that relax these assumptions:

- **Probabilistic planning:** Handles non-deterministic actions with probability distributions over outcomes
- **Conformant planning:** Plans under partial observability without sensing
- **Contingent planning:** Generates conditional plans that include sensing actions
- **Temporal planning:** Reasons about action durations and temporal constraints
- **Multi-agent planning:** Coordinates plans among multiple agents

2.3 The Planning Problem

Given a state transition system $\Sigma = \langle S, A, \gamma \rangle$, we can now formally define the planning problem.

Definizione 2.2 (Planning Problem). A planning problem is a tuple $P = \langle \Sigma, s_0, G \rangle$ where:

- $\Sigma = \langle S, A, \gamma \rangle$ is a state transition system
- $s_0 \in S$ is the **initial state**
- $G \subseteq S$ is a set of **goal states**

Alternatively, goals can be specified as a **goal formula** g , a logical expression that is satisfied by exactly those states in G . This allows more compact and expressive goal specifications.

2.3.1 Goal Specification

Goals can be specified in several ways:

- **Explicit goal states:** Enumerate the set G of acceptable final states. This is impractical for large state spaces.
- **Goal formula:** A logical formula g such that $G = \{s \in S \mid s \models g\}$. For example, in propositional logic: $g = \text{At}(\text{Robot}, \text{RoomB}) \wedge \text{Clean}(\text{RoomB})$
- **Goal conditions:** A set of propositions that must be true in the goal state. This is the most common approach in classical planning.
- **Utility functions:** In optimization settings, goals may be specified as utility functions to maximize or cost functions to minimize.

Definizione 2.3 (Solution Plan). A **solution** to a planning problem $P = \langle \Sigma, s_0, G \rangle$ is a sequence of actions $\pi = \langle a_1, a_2, \dots, a_n \rangle$ such that:

$$s_n = \gamma(\gamma(\dots \gamma(\gamma(s_0, a_1), a_2) \dots, a_{n-1}), a_n) \in G$$

That is, executing the sequence of actions starting from the initial state s_0 results in a goal state.

We can also write this more compactly using the notation $\gamma^*(s, \pi)$ to denote the state reached by executing plan π from state s :

$$\gamma^*(s_0, \pi) \in G$$

2.3.2 Plan Quality

Not all solution plans are equally desirable. Common quality metrics include:

- **Plan length:** Number of actions in the plan. Shorter plans are often preferred.
- **Plan cost:** Sum of action costs. Each action a may have an associated cost $c(a)$.
- **Makespan:** Total execution time (relevant in temporal planning).
- **Resource consumption:** Amount of resources used during execution.

An **optimal plan** minimizes the chosen quality metric among all solution plans.

3 Representation of Planning Problems

3.1 The Challenge of Explicit Representation

While the state transition system formalism is mathematically elegant, it faces a critical practical challenge: **the curse of dimensionality**. Real-world planning domains can have exponentially large state spaces. For example:

- A domain with n boolean variables has 2^n possible states
- A domain with k objects and m binary relations has up to 2^{mk^2} states
- Enumerating all states and transitions explicitly is infeasible for even moderately-sized problems

Esempio 3.1 (Blocks World Complexity). Consider a Blocks World domain with n blocks. The number of possible configurations grows super-exponentially with n . For $n = 10$ blocks, there are more than 10^{13} possible configurations. Explicitly representing the transition function would require storing information about trillions of state-action pairs.

This motivates the need for **compact, structured representations** that exploit regularities in the domain to represent large state spaces and transition functions implicitly.

3.2 Action Schemas: Structured Representation

The classical approach to compact representation uses **action schemas** (also called action templates or operators). An action schema is a parameterized description of a family of related actions.

Definizione 3.1 (Action Schema). An action schema consists of:

- **Name and parameters:** A symbolic name and typed parameters, e.g., $\text{Move}(x, y)$
- **Preconditions:** A logical formula Pre specifying when the action can be executed
- **Effects:** A logical formula Eff specifying how the state changes when the action is executed

Actions are **ground instances** of action schemas, obtained by binding the parameters to specific objects in the domain. For example, the schema $\text{Move}(x, y)$ might generate ground actions $\text{Move}(\text{RoomA}, \text{RoomB})$, $\text{Move}(\text{RoomB}, \text{RoomC})$, etc.

3.2.1 Advantages of Action Schemas

Action schemas provide several benefits:

- **Compactness:** A single schema can represent exponentially many ground actions
- **Generality:** Schemas capture the general structure of actions independent of specific objects
- **Scalability:** Adding new objects to the domain automatically generates new applicable actions
- **Knowledge reuse:** Schemas learned in one domain can be transferred to similar domains

3.3 State Representation: Propositional Logic

In classical planning, states are typically represented using **propositional logic**:

- A state is a set of **atoms** (propositional variables) that are true in that state

- Atoms not in the set are assumed false (**closed-world assumption**)
- Action preconditions and effects are logical formulas over these atoms

This representation allows:

- Compact encoding of structured states
- Efficient reasoning about action applicability and effects
- Use of logical inference techniques for plan generation

3.3.1 State Update Semantics

When an action is executed, the state is updated according to the action's effects:

- **Add list:** Atoms that become true after the action
- **Delete list:** Atoms that become false after the action
- **Frame axioms:** Atoms not mentioned in the effects remain unchanged

The **STRIPS assumption** (named after the Stanford Research Institute Problem Solver) states that only atoms explicitly mentioned in the effects change; all other atoms persist unchanged. This simplifies reasoning about action effects.

3.4 Example: Blocks World

Consider the classic **Blocks World** domain, one of the most studied domains in planning research.

3.4.1 Domain Description

The Blocks World consists of:

- A set of blocks that can be stacked on top of each other
- A table with unlimited space
- A robot arm that can pick up and put down blocks

State representation:

- $\text{On}(x, y)$: Block x is directly on top of block y
- $\text{OnTable}(x)$: Block x is on the table
- $\text{Clear}(x)$: Block x has no blocks on top of it
- $\text{Holding}(x)$: The robot arm is holding block x
- ArmEmpty : The robot arm is not holding anything

3.4.2 Action Schemas

PickUp(x): Pick up block x from the table

Parameters: x (block)

Precondition: $\text{Clear}(x) \wedge \text{OnTable}(x) \wedge \text{ArmEmpty}$

Effect: $\text{Holding}(x) \wedge \neg \text{OnTable}(x) \wedge \neg \text{Clear}(x) \wedge \neg \text{ArmEmpty}$

PutDown(x): Put down block x on the table

Parameters: x (block)

Precondition: $\text{Holding}(x)$

Effect: $\text{OnTable}(x) \wedge \text{Clear}(x) \wedge \text{ArmEmpty} \wedge \neg \text{Holding}(x)$

Stack(x, y): Stack block x on top of block y

Parameters: x, y (blocks)

Precondition: $\text{Holding}(x) \wedge \text{Clear}(y)$

Effect: $\text{On}(x, y) \wedge \text{Clear}(x) \wedge \text{ArmEmpty} \wedge \neg \text{Holding}(x) \wedge \neg \text{Clear}(y)$

Unstack(x, y): Remove block x from on top of block y

Parameters: x, y (blocks)

Precondition: $\text{On}(x, y) \wedge \text{Clear}(x) \wedge \text{ArmEmpty}$

Effect: $\text{Holding}(x) \wedge \text{Clear}(y) \wedge \neg \text{On}(x, y) \wedge \neg \text{Clear}(x) \wedge \neg \text{ArmEmpty}$

3.4.3 Example Problem Instance

Initial state: Blocks A, B, C are all on the table

$$s_0 = \{\text{OnTable}(A), \text{OnTable}(B), \text{OnTable}(C), \text{Clear}(A), \text{Clear}(B), \text{Clear}(C), \text{ArmEmpty}\}$$

Goal: Stack the blocks in the order A on B on C

$$G = \{\text{On}(A, B), \text{On}(B, C), \text{OnTable}(C)\}$$

Solution plan:

1. `PickUp(B)`
2. `Stack(B, C)`
3. `PickUp(A)`
4. `Stack(A, B)`

This plan has length 4 and achieves the goal from the initial state.

Parte II

Advanced Planning Representations

4 Reduction to Propositional Satisfiability

4.1 Motivation

We continue our discussion of the formalization and representation of planning problems. A fundamental technique in automated planning is to **reduce the planning problem to the satisfiability problem of propositional logic**. This reduction allows us to leverage powerful SAT solvers to find plans efficiently.

The key idea is to encode:

- The initial state as a propositional formula
- The goal conditions as a propositional formula
- The action effects and preconditions as propositional formulas
- The constraint that actions must form a valid sequence

If the resulting formula is satisfiable, the satisfying assignment corresponds to a valid plan.

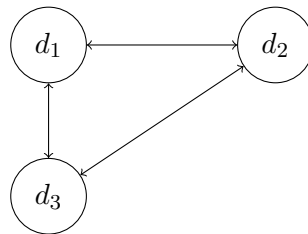
4.2 Example Domain: Dock Worker Robots (DWR)

To illustrate these concepts, we introduce the **Dock Worker Robots (DWR)** domain, a classic planning benchmark that models container logistics at a port.

4.2.1 Domain Description

The DWR world consists of:

- **Docks:** Three docks d_1 , d_2 , d_3 arranged in a triangular configuration
- **Connectivity:** Docks are connected as follows: $d_1 \leftrightarrow d_3$, $d_1 \leftrightarrow d_2$, $d_2 \leftrightarrow d_3$
- **Piles:** Storage locations at docks where containers can be stacked
- **Containers:** Objects that need to be moved between locations
- **Robots:** Mobile agents that can move containers between piles



4.2.2 Initial State Configuration

The initial state of our example problem is:

- **At dock d_1 :** Pile p_1 contains two containers: c_1 (on top) and c_2 (below)
- **At dock d_2 :**

- Pile p_2 contains one container: c_3
- Pile p_3 is empty
- **At dock d_3 :** Pile p_4 is empty

4.3 Formal Representation

4.3.1 Objects and Types

The domain contains objects of different **sorts** (types). Each object is represented by a constant symbol with an associated type:

Docks:

$$\text{Docks} = \{d_1, d_2, d_3\}$$

where each d_i is a constant symbol of sort **Dock**.

Containers:

$$\text{Containers} = \{c_1, c_2, c_3\}$$

where each c_i is a constant symbol of sort **Container**.

Robots:

$$\text{Robots} = \{r_1, r_2\}$$

where each r_i is a constant symbol of sort **Robot**.

Piles:

$$\text{Piles} = \{p_1, p_2, p_3, p_4\}$$

where each p_i is a constant symbol of sort **Pile**.

The complete set of objects is:

$$\text{Objects} = \text{Docks} \cup \text{Piles} \cup \text{Containers} \cup \text{Robots}$$

4.3.2 Static Relations

Some relations in the domain are **static**—they do not change during plan execution. These can be specified once and remain constant throughout.

Adjacency relation: Specifies which docks are directly connected, allowing robots to travel between them.

$$\text{adjacent} = \{(d_1, d_3), (d_3, d_1), (d_2, d_3), (d_3, d_2), (d_1, d_2), (d_2, d_1)\}$$

This relation is symmetric: if dock d_i is adjacent to dock d_j , then d_j is adjacent to d_i .

Pile-Dock association: Each pile is permanently located at a specific dock.

$$\text{at}(p_1, d_1)$$

$$\text{at}(p_2, d_2)$$

$$\text{at}(p_3, d_2)$$

$$\text{at}(p_4, d_3)$$

4.3.3 Rigid Relations

The adjacency relation is defined over $\text{Docks} \times \text{Docks}$. This is called a **rigid relation** (also known as a **static relation**)—it does not change during plan execution. Actions cannot modify the physical layout of the docks or their connectivity.

Rigid relations are important because:

- They reduce the state space by factoring out unchanging aspects
- They can be checked once at planning time rather than during execution
- They simplify action preconditions by providing fixed background knowledge

4.4 State Representation in DWR

4.4.1 What is a State?

A **state** is an assignment of values to propositional variables. More precisely, a state can be represented in two equivalent ways:

1. **First-order ground atoms:** Predicates where all arguments are constants (no variables)
2. **First-order ground terms:** Terms where all arguments are constants

Important restriction: We only allow **flat expressions**, meaning expressions with depth 1 (no nesting).

Esempio 4.1 (Forbidden Nesting). Consider the predicate $\text{loc}(x, y)$ meaning "the location of x is y ". We do **not** allow nested expressions such as:

$$\text{loc}(r_1, \text{loc}(r_2))$$

This would represent "the location of r_1 is the location of r_2 ", which violates the flatness constraint.

All atoms and terms must be flat—arguments of predicates and functions must be constants, not complex expressions.

4.4.2 State as Truth Assignment

A state assigns truth values to ground atoms. Consider describing the initial state s_0 of our DWR example:

$$\begin{aligned} &\text{loc}(r_1, d_1) \\ &\text{pos}(p_1, d_1) \\ &\text{pos}(c_2, p_1) \end{aligned}$$

These atoms being listed in the state means they are **implicitly assigned true**:

$$\begin{aligned} \text{loc}(r_1, d_1) &\leftarrow \text{true} \\ \text{pos}(p_1, d_1) &\leftarrow \text{true} \\ \text{pos}(c_2, p_1) &\leftarrow \text{true} \end{aligned}$$

Atoms not mentioned in the state are assumed false (closed-world assumption).

4.5 Two Representation Formalisms

There are two common approaches to representing planning problems, differing in how they encode state information.

4.5.1 Classical (Propositional) Representation

In the **classical representation**, states are represented using only **Boolean assignments** to propositional atoms. Each possible fact about the world is represented as a Boolean variable.

Example atoms:

- $\text{loc}(r_1, d_1)$: Robot r_1 is at dock d_1 (true/false)
- $\text{pos}(c_2, p_1)$: Container c_2 is at pile p_1 (true/false)
- $\text{empty}(r_1)$: Robot r_1 is not carrying anything (true/false)

Characteristics:

- Simple and uniform representation
- Direct correspondence to propositional logic
- Easy to encode as SAT problems
- May require many atoms for complex domains

4.5.2 State Variable Representation

The **state variable representation** uses a slightly different syntax based on **function symbols**. Instead of Boolean atoms, we use variables that can take on different values from a finite domain.

Example:

$$\text{loc}(r_1) = d_1$$

This means "the location of robot r_1 is d_1 ". Here, $\text{loc}(r_1)$ is a **state variable** (a function that returns a value), and d_1 is its current value.

Comparison with classical representation:

Classical	State Variable
$\text{loc}(r_1, d_1) = \text{true}$	$\text{loc}(r_1) = d_1$
$\text{loc}(r_1, d_2) = \text{false}$	
$\text{loc}(r_1, d_3) = \text{false}$	

Advantages of state variables:

- More compact: one variable instead of multiple Boolean atoms
- Explicitly represents mutual exclusion (robot can only be at one location)
- Natural for domains with multi-valued attributes
- Reduces the number of variables in SAT encodings

Relationship: The two representations are **equivalent in expressive power**. Any problem representable in one formalism can be translated to the other. The choice depends on:

- The planning algorithm being used
- The structure of the domain

- Efficiency considerations for the specific problem

4.5.3 Example: Complete State Representation

Let us represent the complete initial state s_0 of our DWR example in both formalisms.

Classical representation (partial):

```

loc( $r_1, d_1$ ) = true
loc( $r_2, d_2$ ) = true
pos( $c_1, p_1$ ) = true
pos( $c_2, p_1$ ) = true
pos( $c_3, p_2$ ) = true
on( $c_1, c_2$ ) = true
empty( $r_1$ ) = true
empty( $r_2$ ) = true
top( $c_1, p_1$ ) = true
top( $c_3, p_2$ ) = true

```

State variable representation:

```

loc( $r_1$ ) =  $d_1$ 
loc( $r_2$ ) =  $d_2$ 
pos( $c_1$ ) =  $p_1$ 
pos( $c_2$ ) =  $p_1$ 
pos( $c_3$ ) =  $p_2$ 
on( $c_1$ ) =  $c_2$ 
loaded( $r_1$ ) = nil
loaded( $r_2$ ) = nil
top( $p_1$ ) =  $c_1$ 
top( $p_2$ ) =  $c_3$ 

```

Both representations capture the same information but organize it differently. The state variable representation is more compact and explicitly enforces constraints (e.g., a robot can only be at one location).

4.6 Key Insight: States as Boolean Assignments

The fundamental insight is that regardless of the representation formalism chosen:

A state is fundamentally a Boolean assignment

Whether we use propositional atoms or state variables, we are ultimately assigning truth values (or equivalently, values from finite domains) to a set of variables.

5 Action Schemas in DWR

5.1 Structure of Action Schemas

An **action schema** α is a template for generating ground actions. It consists of:

Definizione 5.1 (Action Schema). An action schema α has the following components:

- **Name and parameters:** $\alpha : \text{name}(z_1, z_2, \dots, z_k)$ where z_i are typed parameters
- **Preconditions:** $\text{Pre}(\alpha) = \{p_1, p_2, \dots, p_n\}$ where each p_i is a literal

- **Effects:** $\text{Eff}(\alpha) = \{q_1, q_2, \dots, q_m\}$ where each q_i is a literal
- **Cost:** $c(\alpha) \in \mathbb{N}_0$ (typically a non-negative integer)

Important constraint: The parameters z_i of α can appear in the precondition literals p_j and effect literals q_j . The arguments in these literals can be either:

- **Constants:** Specific objects from the domain
- **Parameters:** Variables that will be instantiated

No arbitrary free variables are allowed—all variables must be parameters of the action schema.

Definizione 5.2 (Ground Action). A **ground action** a is obtained by substituting all parameters z_i in action schema α with specific constants from the domain. The action a is an **instance** of α .

5.2 Example: Take/Load Action

Consider the action of a robot taking a container from a pile. We define the action schema:

$$\alpha : \text{take}(r, c, c', p, d)$$

Parameters:

- r : robot (type `Robot`)
- c : container to take (type `Container`)
- c' : container below c (type `Container` $\cup$ `{nil}`)
- p : pile (type `Pile`)
- d : dock (type `Dock`)

5.2.1 Correcting Static Relations

Before defining the preconditions, we note that there are **two rigid (static) relations** in the DWR domain:

1. `adjacent : Docks  $\times$  Docks` — specifies which docks are connected
2. `at : Piles  $\times$  Docks` — specifies which pile is located at which dock

Both relations are rigid because the physical layout of the port does not change during plan execution.

5.2.2 Preconditions

For the `take` action to be executable, the following conditions must hold:

$$\text{Pre}(\text{take}(r, c, c', p, d)) = \left\{ \begin{array}{l} \text{loc}(r) = d \\ \text{at}(p, d) \\ \text{pos}(c) = p \\ \text{on}(c) = c' \\ \text{top}(p) = c \\ \text{cargo}(r) = \text{nil} \end{array} \right\}$$

Interpretation:

- The robot r is at dock d
- The pile p is located at dock d (static relation)
- Container c is positioned at pile p
- Container c is on top of container c' (or c' is `nil` if c is at the bottom)
- Container c is the top container of pile p
- The robot r is not carrying anything

5.2.3 Effects

When the `take` action is executed, the following changes occur:

$$\text{Eff}(\text{take}(r, c, c', p, d)) = \left\{ \begin{array}{l} \text{cargo}(r) = c \\ \text{top}(p) = c' \\ \text{pos}(c) = r \end{array} \right\}$$

Interpretation:

- The robot r is now carrying container c
- The top of pile p is now c' (the container that was below c)
- The position of container c is now the robot r

5.3 Action Applicability

Definizione 5.3 (Action Applicability). A ground action a is **applicable** (or **executable**) in state s if all preconditions of a are satisfied in s :

$$a \text{ applicable in } s \iff s \models \text{Pre}(a)$$

5.3.1 Examples

Consider two ground actions:

Example 1: $a_1 = \text{take}(r_1, c_1, c_2, p_1, d_1)$

In the initial state s_0 where:

- $\text{loc}(r_1) = d_1$
- $\text{at}(p_1, d_1)$
- $\text{pos}(c_1) = p_1$
- $\text{on}(c_1) = c_2$
- $\text{top}(p_1) = c_1$
- $\text{cargo}(r_1) = \text{nil}$

All preconditions are satisfied, so a_1 is **executable/applicable** in s_0 .

Example 2: $a_2 = \text{take}(r_2, c_3, \text{nil}, p_4, d_3)$

This action requires:

- $\text{loc}(r_2) = d_3$

- $\text{at}(p_4, d_3)$
- $\text{pos}(c_3) = p_4$
- $\text{top}(p_4) = c_3$

However, in s_0 , we have $\text{pos}(c_3) = p_2$ (not p_4) and $\text{loc}(r_2) = d_2$ (not d_3). Therefore, a_2 is **not executable/not applicable** in s_0 .

6 State Transition Function

6.1 Formal Definition

The **state transition function** γ defines how actions transform states:

$$\gamma : \text{State} \times \text{Action} \rightarrow \text{State}$$

where $\gamma(s, a)$ computes the successor state obtained by executing ground action a in state s .

6.2 Computing the Successor State

Given a state s and a ground action a (which is a ground instance of an action schema α), the successor state $\gamma(s, a) = s'$ is defined if and only if:

1. $s \models p$ for all $p \in \text{Pre}(a)$ (all preconditions are satisfied)
2. s' is consistent (the resulting state contains no contradictions)

The successor state s' is computed as:

$$s' = \{q \mid q \in \text{Eff}(a)\} \cup \{q \mid s \models q, \neg q \notin \text{Eff}(a), q \notin \text{Eff}(a)\}$$

Interpretation: The new state consists of:

- All literals in the effects of a
- All literals from s that are not affected by a (neither the literal nor its negation appears in the effects)

6.3 Goal States

While the initial state is a specific assignment, goals are typically specified more flexibly.

Definizione 6.1 (Goal Formula). A **goal formula** G is a conjunction of literals specifying desired properties of goal states.

Definizione 6.2 (Goal States). The set of **goal states** is:

$$S_g = \{s \mid s \models G\}$$

That is, all state assignments that satisfy the goal formula G .

6.4 Comparing Representations: Simplified DWR Example

Let us compare the classical and state variable representations using a simplified version of the DWR domain with three action schemas.

6.4.1 Classical Representation

In the classical representation, we use propositional atoms to represent facts:

Action 1: $\text{take}(r, c, l)$ — robot r takes container c at location l

$$\begin{aligned} \text{Pre: } & \text{loc}(r, l) \wedge \text{pos}(c, l) \wedge \neg \text{loaded}(r) \\ \text{Eff: } & \text{pos}(c, r) \wedge \neg \text{pos}(c, l) \wedge \text{loaded}(r) \end{aligned}$$

Action 2: $\text{move}(r, l, m)$ — robot r moves from location l to location m

$$\begin{aligned} \text{Pre: } & \text{loc}(r, l) \\ \text{Eff: } & \text{loc}(r, m) \wedge \neg \text{loc}(r, l) \end{aligned}$$

Action 3: $\text{put}(r, c, l)$ — robot r puts down container c at location l

$$\begin{aligned} \text{Pre: } & \text{loc}(r, l) \wedge \text{pos}(c, r) \\ \text{Eff: } & \text{pos}(c, l) \wedge \neg \text{pos}(c, r) \wedge \neg \text{loaded}(r) \end{aligned}$$

6.4.2 State Variable Representation

In the state variable representation, we use functional notation with assignments:

Action 1: $\text{take}(r, c, l)$

$$\begin{aligned} \text{Pre: } & \text{loc}(r) = l \wedge \text{pos}(c) = l \wedge \neg \text{loaded}(r) \\ \text{Eff: } & \text{pos}(c) \leftarrow r \wedge \text{loaded}(r) \end{aligned}$$

Action 2: $\text{move}(r, l, m)$

$$\begin{aligned} \text{Pre: } & \text{loc}(r) = l \\ \text{Eff: } & \text{loc}(r) \leftarrow m \end{aligned}$$

Action 3: $\text{put}(r, c, l)$

$$\begin{aligned} \text{Pre: } & \text{loc}(r) = l \wedge \text{pos}(c) = r \\ \text{Eff: } & \text{pos}(c) \leftarrow l \wedge \neg \text{loaded}(r) \end{aligned}$$

6.4.3 Key Differences

- **Classical:** Requires explicit deletion of old values (e.g., $\neg \text{loc}(r, l)$ when moving)
- **State Variable:** Implicit deletion through reassignment (e.g., $\text{loc}(r) \leftarrow m$ automatically removes the old location)
- **Compactness:** State variables are more concise for multi-valued attributes

7 Planning as Satisfiability (SAT)

7.1 Recap: Core Concepts

Before introducing the SAT encoding, let us recap the fundamental concepts:

- **State:** A Boolean assignment to propositional (ground) literals

- **Action:** Characterized by:
 - **Preconditions:** A set (or conjunction) of literals
 - **Effects:** A set (or conjunction) of literals

7.2 Temporal Instantiation of Variables

To encode planning problems as SAT, we introduce a temporal dimension to our propositional variables.

Definizione 7.1 (Temporal Instantiation). Let X be the set of propositional variables describing the domain. The **temporal instantiation** of X at time t is:

$$X@t = \{x@t \mid x \in X\} \quad \text{for } t \geq 0$$

where $x@t$ is a propositional variable representing the truth value of x at time step t .

Intuition: Since a plan is a sequence of actions $(a_0, a_1, a_2, \dots, a_{n-1})$ of length n , we need to track the truth value of each proposition at each time step from 0 to n .

7.3 The Bounded Planning Problem

Definizione 7.2 (Bounded Planning Problem). Given a planning problem $P = \langle \Sigma, I, G \rangle$, the **bounded planning problem** asks:

Does there exist a plan solving P with length exactly n ?

This is a decision problem parameterized by the plan length n .

7.4 SAT Encoding

The bounded planning problem can be encoded as a propositional satisfiability problem.

Teorema 7.1 (Planning as SAT). *For a planning problem P and bound n , there exists a propositional formula Φ_n such that:*

$$\Phi_n \text{ is satisfiable} \quad \Leftrightarrow \quad \text{there exists a plan of length } n \text{ for } P$$

The formula Φ_n encodes:

1. **Initial state:** $I@0$ (the initial state holds at time 0)
2. **Goal state:** $G@n$ (the goal holds at time n)
3. **Action preconditions and effects:** For each time step $t \in \{0, 1, \dots, n-1\}$, exactly one action is executed and its effects are correctly applied
4. **Frame axioms:** Variables not affected by actions remain unchanged

7.5 Solving Planning via Iterative Deepening

Since we don't know the optimal plan length in advance, we use **iterative deepening**:

Algorithm 1 Planning via SAT with Iterative Deepening

```
1:  $n \leftarrow 1$ 
2: while true do
3:   Construct formula  $\Phi_n$ 
4:   if  $\Phi_n$  is satisfiable then
5:     Extract plan from satisfying assignment
6:     return plan
7:   end if
8:    $n \leftarrow n + 1$ 
9: end while
```

Process:

1. Try $n = 1$: Check if Φ_1 is satisfiable
2. If not, try $n = 2$: Check if Φ_2 is satisfiable
3. If not, try $n = 3$: Check if Φ_3 is satisfiable
4. Continue until a satisfiable formula is found

Advantages:

- Finds shortest plans (optimal in number of steps)
- Leverages highly optimized SAT solvers
- Scales well for many planning domains

Challenges:

- Formula size grows with n
- May need to try many values of n before finding a solution
- Encoding frame axioms can be expensive

Parte III

SAT Solvers and Planning

8 The Boolean Satisfiability Problem

8.1 Definition and Scope

The **Boolean Satisfiability Problem (SAT)** is one of the most fundamental problems in computer science and forms the foundation for many automated reasoning techniques, including planning.

Definizione 8.1 (SAT Problem). The SAT problem asks: Given a propositional formula, does there exist an interpretation (truth assignment) that makes the formula true?

Formally, given a set of clauses S in propositional logic, determine whether S is satisfiable.

8.2 Propositional Logic Basics

To understand SAT, we must first establish the basic elements of propositional logic:

- **Propositional variables:** Let X be a finite set of propositional variables, where each variable can assume one of two values: **true** (1) or **false** (0)
- **Atom (Atomic formula):** A single propositional variable $x \in X$
- **Literal:** Either an atom x or its negation $\neg x$
- **Clause:** A disjunction of literals: $L_1 \vee L_2 \vee \dots \vee L_n$ where each L_i is a literal

8.3 Interpretations and Truth Assignments

Definizione 8.2 (Interpretation). An **interpretation** (or **truth assignment**) is a function:

$$v : X \rightarrow \{0, 1\}$$

that assigns a Boolean value to each propositional variable.

Number of interpretations: If $|X| = n$ (i.e., we have n propositional variables), then there are exactly 2^n possible interpretations.

8.3.1 Satisfiability of Clauses

Definizione 8.3 (Clause Satisfaction). A clause $C = L_1 \vee L_2 \vee \dots \vee L_n$ is **satisfied** by an interpretation v if there exists at least one literal L_i such that $v(L_i) = \text{true}$ for some $i \in \{1, 2, \dots, n\}$.

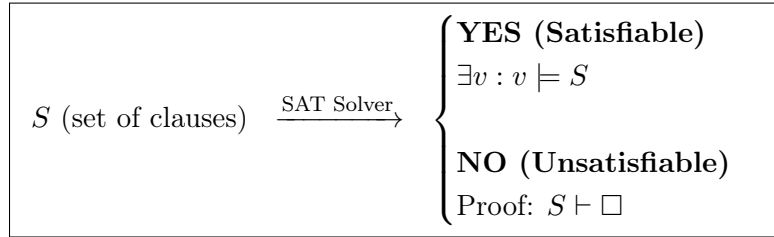
We write: $v \models C$

Definizione 8.4 (Formula Satisfaction). A set of clauses $S = \{C_1, C_2, \dots, C_m\}$ is **satisfiable** if there exists an interpretation v that satisfies all clauses in S :

$$v \models S \quad \Leftrightarrow \quad v \models C_i \text{ for all } i \in \{1, 2, \dots, m\}$$

8.4 SAT Decision Procedure

The SAT problem can be viewed as a decision procedure with the following input-output specification:



Output cases:

1. **YES (Satisfiable):** There exists an interpretation v such that $v \models S$. The SAT solver typically returns this satisfying assignment.
2. **NO (Unsatisfiable):** No interpretation satisfies S . The solver may provide a proof of unsatisfiability, showing that S derives a contradiction (denoted $S \vdash \square$, where $\square$ represents the empty clause).

8.5 The Empty Clause

Definizione 8.5 (Empty Clause). The **empty clause**, denoted $\square$ (or $\Box$), is a clause with no literals. It is **false in all interpretations** and therefore represents a contradiction.

Significance: If a SAT solver derives the empty clause from the input formula S , this constitutes a proof that S is unsatisfiable, since deriving a contradiction means no interpretation can satisfy all clauses.

9 Planning Problem Formalization

We formalized the planning problem with:

$$P = \langle \Sigma, s_0, S_g \rangle$$

where $\Sigma = \langle S, A, \gamma \rangle$.

- s_0 is the initial state
- S_g is the set of goal states: $S_g = \{s \mid s \models G\}$
- G is the goal formula (a conjunction of atoms)

9.1 States as Interpretations

States correspond to interpretations in propositional logic:

- $\text{at}(r_1, l_1)$: ground positive literal
- $\neg \text{loaded}(r_1)$: ground negative literal

Abstraction: From ground first-order literals to propositional literals. With finitely many ground atoms (literals) in the planning problem, we obtain finitely many variables $|X| = n$.

10 Bounded Planning Problem

Definizione 10.1 (Bounded Planning Problem). Does there exist a plan π that solves planning problem P and has length k (i.e., $|\pi| = k$, where π is a sequence of k actions)?

10.1 Iterative Deepening on Plan Length

We use iterative deepening to find the shortest plan by increasing the bound k :

$$P \xrightarrow{\text{translation}} \Phi \xrightarrow{\text{SAT}} \begin{cases} \text{YES : interpretation } v \models \Phi \\ \text{encodes plan } \pi \text{ of length } k \\ \text{NO : no plan of length } k \text{ solves } P \end{cases}$$

The process iterates: $k := k + 1$ until a solution is found.

Key insight: The SAT encoding Φ includes:

- Initial state constraints at time 0
- Goal constraints at time k
- Action preconditions and effects for each time step
- Frame axioms ensuring persistence of unchanged literals

10.2 Temporal Encoding of Literals

A key aspect of the SAT encoding for planning is the **temporal instantiation** of literals. Since a plan of length k requires tracking state changes over time, each literal is indexed by time steps.

- **Plan length k** means we have $0 \leq i \leq k - 1$ time steps (transitions)
- Each literal f becomes $f@t$ where t ranges from 0 to k
- State i is the state after executing i actions (before the i -th action)

10.2.1 Example: Temporal Instantiation

Consider the literal $\text{at}(r_1, l_1)$ ("robot r_1 is at location l_1 "):

$$\text{at}(r_1, l_1) \xrightarrow{\text{temporal encoding}} \text{at}(r_1, l_1, i)$$

Similarly, the negative literal $\neg \text{loaded}(r_1)$ becomes $\neg \text{loaded}(r_1, i)$, meaning "robot r_1 is not loaded at stage i ".

Definizione 10.2 (State Transition). Step i (where $0 \leq i \leq k - 1$) takes us from state i to state $i + 1$ by executing one action.

10.3 Encoding the Initial State

The initial state s_0 is encoded as a conjunction of constraints at time $t = 0$:

$$\Phi_{s_0} = \bigwedge_{f \in s_0} f@0 \quad \wedge \quad \bigwedge_{f \notin s_0} \neg f@0$$

Where f represents ground literals.

10.3.1 From Literals to Propositional Variables

Each temporally indexed literal corresponds to a propositional variable:

- $\text{at}(r_1, l_1, 0)$ corresponds to variable $x_{\text{at}(r_1, l_1)}@0$
- $\neg\text{loaded}(r_1, 0)$ corresponds to $\neg x_{\text{loaded}(r_1)}@0$

10.3.2 Example: Complete Initial State Encoding

Consider an initial state containing $\text{at}(r_1, l_1)$ but not $\text{loaded}(r_1)$:

$$\begin{array}{lll} \text{Literals in } s_0 : & \text{at}(r_1, l_1) & \wedge \neg\text{loaded}(r_1) \wedge \dots \\ \text{Temporal encoding:} & \text{at}(r_1, l_1, 0) & \wedge \neg\text{loaded}(r_1, 0) \wedge \dots \\ \text{SAT variables:} & x_{\text{at}(r_1, l_1)}@0 & \wedge \neg x_{\text{loaded}(r_1)}@0 \wedge \dots \end{array}$$

For literals not in the initial state (e.g., $\text{at}(r_1, l_2)$, $\text{at}(r_1, l_3)$), these are assigned false:

$$\neg\text{at}(r_1, l_2, 0), \quad \neg\text{at}(r_1, l_3, 0)$$

Alternative notation: The initial state encoding can also be written using set notation:

$$\Phi_{s_0} = \bigwedge_{x \in S_0^+} x_0 \wedge \bigwedge_{x \in S_0^-} \neg x_0$$

where S_0^+ is the set of positive literals in s_0 and S_0^- is the set of negative literals in s_0 .

10.4 Encoding the Goal States

The goal conditions must be satisfied at the final time step k .

Definizione 10.3 (Goal Encoding). The goal states are encoded at time k as:

$$\Phi_G = \bigwedge_{x \in G^+} x_k \wedge \bigwedge_{x \in G^-} \neg x_k$$

where G^+ is the set of positive literals in the goal G and G^- is the set of negative literals in G .

Interpretation: This formula ensures that all positive goal conditions are true and all negative goal conditions are false in the final state after executing k actions.

10.5 Encoding Actions

The planning problem is defined as $\Sigma = \langle S, A, \gamma \rangle$, where A is the set of all ground instances of the action schemas.

10.5.1 Action Preconditions

For each action $a \in A$ at each time step i (where $0 \leq i \leq k-1$), we encode the constraint that if the action is executed, its preconditions must hold:

Definizione 10.4 (Action Precondition Encoding).

$$\forall a \in A, \forall i \in \{0, \dots, k-1\} : \quad a_i \rightarrow \bigwedge_{p \in \text{prec}(a)} p_i$$

where $\text{prec}(a)$ denotes the set of preconditions of action a .

Interpretation: This formula states that if action a is executed at time i , then all of its preconditions must be satisfied at time i . The implication can be rewritten in CNF as:

$$\neg a_i \vee \bigwedge_{p \in \text{prec}(a)} p_i$$

10.5.2 Mutual Exclusion of Actions

At each time step, at most one action can be executed. This constraint is encoded as follows:

Definizione 10.5 (Action Mutual Exclusion). For all pairs of distinct actions $(a, b) \in A \times A$ where $a \neq b$, and for all time steps i where $0 \leq i \leq k - 1$:

$$\neg a_i \vee \neg b_i$$

This formula ensures that actions a and b cannot both be executed at the same time step i .

Equivalent formulations:

- $a_i \rightarrow \neg b_i$: If action a is executed at time i , then action b is not executed
- $b_i \rightarrow \neg a_i$: If action b is executed at time i , then action a is not executed

10.6 Complete SAT Encoding

The complete SAT encoding Φ for a bounded planning problem with plan length k is the conjunction of the following sets of propositional formulas:

$$\Phi = \Phi_{s_0} \wedge \Phi_G \wedge \Phi_{\text{actions}} \wedge \Phi_{\text{mutex}} \wedge \Phi_{\text{frame}}$$

where:

- Φ_{s_0} : Initial state constraints
- Φ_G : Goal state constraints
- Φ_{actions} : Action precondition and effect constraints
- Φ_{mutex} : Mutual exclusion constraints between actions
- Φ_{frame} : Frame axioms (persistence of unchanged literals)

10.7 Plan Extraction from SAT Model

Once the SAT solver returns a satisfying assignment, we can extract the corresponding plan.

Definizione 10.6 (Plan Extraction). Let v be a satisfying interpretation (model) returned by the SAT solver, represented as a set of literals. The plan π is extracted as follows:

$$\pi = \langle a_0, a_1, \dots, a_{k-1} \rangle$$

where a_i is the unique action such that $v \models a_i$ (i.e., a_i is assigned **true** in the model v) for each time step $i \in \{0, 1, \dots, k - 1\}$.

Extraction procedure:

1. Filter the model v to obtain only the action literals: $\{a_i \mid a \in A, 0 \leq i \leq k - 1, v \models a_i\}$

2. For each time step i , identify the action a such that a_i is true in v
3. Construct the sequence $\pi = \langle a_0, a_1, \dots, a_{k-1} \rangle$

Nota. Due to the mutual exclusion constraints, at most one action is true at each time step. If no action is true at some time step i , this corresponds to a "no-op" (no operation) action.